

PIZZA SALES ANALYSIS

SQL Portfolio Project

Business Insights from 2015 Sales Data

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Tool: MySQL 8.0+ | Focus: SQL & Business Analytics

1. Project Overview

This project analyzes transactional pizza sales data to understand order volume, revenue performance, product demand, category contribution and time-based ordering patterns. The analysis is designed as a practical Data Analyst portfolio project using relational SQL.

Objectives

- Measure total orders and total revenue.
- Identify high-performing pizzas, sizes and categories.
- Analyze order activity by hour of day.
- Calculate daily sales metrics and revenue contribution.
- Apply advanced SQL concepts such as window functions for cumulative and ranked analysis.

2. Dataset & Database Structure

The dataset contains order-level and product-level information across four related tables. The final SQL schema uses primary and foreign keys to preserve the relational structure.

Table	Rows	Purpose
orders	21,350	Order date and time
order_details	48,620	Pizza quantity by order
pizzas	96	Pizza size and price
pizza_types	32	Pizza name, category and ingredients

Data validation found no missing values or duplicate rows in the four supplied CSV datasets.

3. Key Performance Indicators

Metric	Result
Total Orders	21,350
Total Revenue	\$817,860.05
Average Pizzas per Day	138.47
Highest-Priced Pizza	The Greek Pizza — \$35.95
Most Ordered Size	L
Top Quantity Category	Classic

4. Product Performance

Top 5 pizza types by quantity ordered:

Rank	Pizza Type	Quantity
1	The Classic Deluxe Pizza	2,453
2	The Barbecue Chicken Pizza	2,432
3	The Hawaiian Pizza	2,422
4	The Pepperoni Pizza	2,418
5	The Thai Chicken Pizza	2,371

Top 3 pizza types by revenue:

Rank	Pizza Type	Revenue
1	The Thai Chicken Pizza	\$43,434.25
2	The Barbecue Chicken Pizza	\$42,768.00
3	The California Chicken Pizza	\$41,409.50

5. Category & Size Analysis

Category	Quantity Ordered	Revenue Share
Classic	14,888	26.91%
Supreme	11,987	25.46%
Veggie	11,649	23.68%
Chicken	11,050	23.96%

Pizza size quantity distribution:

Size	Quantity
L	18,956
M	15,635
S	14,403
XL	552
XXL	28

6. Time-Based Ordering Analysis

Orders are concentrated around lunch and evening periods. The highest observed order volume occurs at 12:00 with 2,520 orders, followed by 13:00 with 2,455.

Hour	Orders
09:00	1
10:00	8
11:00	1,231
12:00	2,520
13:00	2,455
14:00	1,472
15:00	1,468
16:00	1,920
17:00	2,336
18:00	2,399
19:00	2,009
20:00	1,642
21:00	1,198
22:00	663
23:00	28

7. SQL Techniques Demonstrated

- Relational joins across four connected tables.
- Aggregation using COUNT, SUM and AVG.
- GROUP BY and ORDER BY for business summaries.
- Subquery for total-revenue comparison.
- Date/time extraction using HOUR().
- Window functions for cumulative revenue.
- DENSE_RANK with PARTITION BY for top-3 analysis within categories.

Interview-ready project summary

"I built a MySQL pizza sales analysis project using four relational tables. I used joins and aggregations to calculate revenue, order volume and product performance, then applied window functions to analyze cumulative revenue and rank the top-performing pizzas within each category."

8. Business Insights

- Large size pizzas lead quantity demand with 18,956 pizzas ordered.
- The Classic category has the highest quantity at 14,888.
- The top revenue-generating pizza is The Thai Chicken Pizza, contributing \$43,434.25.
- The busiest individual hour is 12:00, indicating a strong lunch-time demand window.
- Revenue is distributed across all four categories, with Classic having the largest share.
- The results can support inventory planning, staffing decisions and product-promotion strategies.

Portfolio Note

This report is prepared for portfolio and interview demonstration purposes. The analysis is based on the supplied 2015 sales dataset and should not be interpreted as current real-world sales information.

9. Project File Guide

File	Purpose
01_database_setup.sql	Creates the database, tables, keys and relationships.
02_data_import.sql	Imports all four CSV datasets using LOCAL INFILE.
03_analysis_queries.sql	Contains 13 business-focused SQL analysis queries.
orders.csv	Order-level date and time data.
order_details.csv	Order line and quantity data.
pizzas.csv	Pizza size and price master data.
pizza_types.csv	Pizza names, categories and ingredients.
README.md	GitHub documentation and project explanation.
LinkedIn_Post.txt	Ready-to-use LinkedIn post caption.

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